

$$A \in \mathbb{R}^{m \times m} \quad x \in \mathbb{R}^m \quad b \in \mathbb{R}^m$$

$$Ax = b \xrightarrow[\text{elimination}]{\text{TRANSFORMAZIONE}} Ux = c \xrightarrow[\text{back substitution}]{\text{RISOLUZIONE}} x$$

di
Gauss

$O\left(\frac{m^3}{3}\right)$ $O\left(\frac{m^2}{2}\right)$

COSTO COMPUTAZIONALE DEL METODO DI SOSTITUZIONE ALL'INDIETRO

$$\begin{cases} u_{11}x_1 + u_{12}x_2 + \dots + u_{1m}x_m = b_1 \\ \vdots \\ u_{i1}x_1 + u_{i2}x_2 + \dots + u_{im}x_m = b_i \\ \vdots \\ u_{m-1,1}x_1 + u_{m-1,2}x_2 + \dots + u_{m-1,m}x_m = b_{m-1} \\ u_{m1}x_1 + u_{m2}x_2 + \dots + u_{mm}x_m = b_m \end{cases}$$

$\sum_{i=1}^m i = \frac{n(n+1)}{2}$

proprietà nota

M = moltiplicazioni/divisioni

$$\begin{array}{r} mM + 1 + \\ \text{(divisioni)} \quad 2 + \\ \quad \quad \quad 3 + \\ \quad \quad \quad \vdots \\ \quad \quad \quad m-1 \\ \hline \sum_{i=1}^{m-1} i \end{array}$$

$$\begin{aligned} &= m + \sum_{i=1}^{m-1} i = m + \frac{(m-1) \cdot m}{2} \\ &= \frac{2m + m^2 - m}{2} = \frac{m^2 + m}{2} \\ &= \frac{m(m+1)}{2} = \frac{m^2}{2} + \frac{m}{2} = O\left(\frac{m^2}{2}\right) \end{aligned}$$

FORMULE DEL METODO DI ELIMINAZIONE DI GAUSS

AL GENERICO passo k

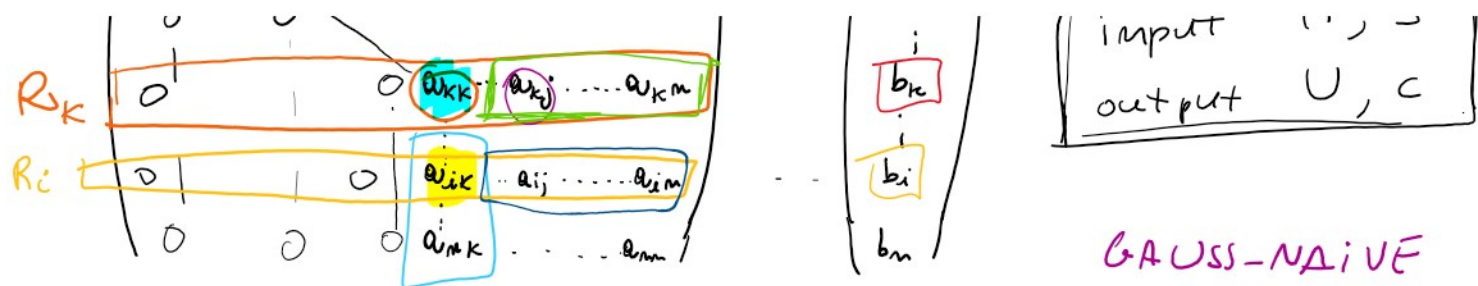
$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ 0 & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{km} \end{pmatrix}$$

colonna k

R_k

$$\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_k \end{pmatrix}$$

input A, b
output U, c



des annullare a_{ik} per $i = k+1, \dots, m$ utilizzando $a_{kk} \neq 0$

for $k=1 \dots m-1$ if $a_{kk} = 0$ return

for $i=k+1 \dots m$

$$m = \frac{a_{ik}}{a_{kk}}$$

$$a_{ik} = 0$$

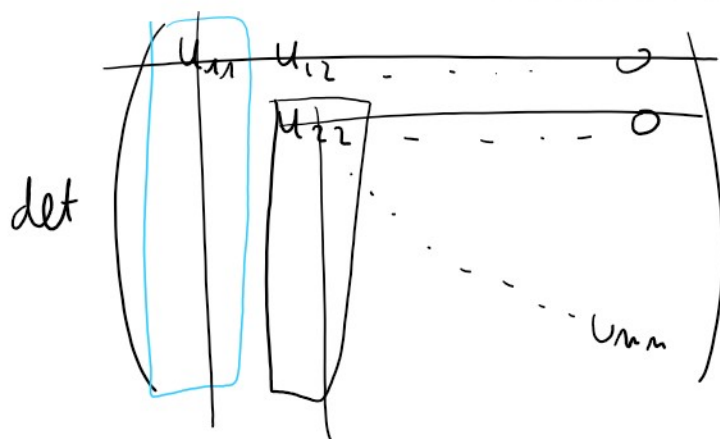
for $j=k+1 \dots m$

$$a_{ij} = a_{ij} - m \cdot a_{kj}; \quad b_i = b_i - m \cdot b_k$$

end

end

$$A(i, k+1:n) = A(i, k+1:n) - m \cdot A(k, k+1:n);$$



$$= u_{11} \cdot u_{22} \cdot \dots \cdot u_{nn} = \prod_{i=1}^n u_{ii}$$

$$\det(A) \neq 0$$

$$A = \begin{pmatrix} 0 & 2 \\ 1 & 4 \end{pmatrix}$$

$$\det(A) = 0 \cdot 4 - 2 \cdot 1 = -2$$

```

gaussNaiveOld.m
1 function [A,b,deter] = gaussNaiveOld(A,b)
2 deter=1; n=size(A,1);
3 for k=1:n-1
4     if (A(k,k)==0)
5         deter=[]; disp('Metodo non applicabile');
6         return
7     end
8     deter=deter*A(k,k);
9     for i=k+1:n
10        m=A(i,k)/A(k,k);
11        A(i,k)=0;

```

```

gaussNaiveOld.m x +
1 function [A,b,deter] = gaussNaiveOld(A,b)
2     deter=1; n=size(A,1);
3     for k=1:n-1
4         if ( A(k,k)==0 )
5             deter=[]; disp('Metodo non applicabile');
6             return
7         end
8         deter=deter*A(k,k);
9         for i=k+1:n
10            m=A(i,k)/A(k,k);
11            A(i,k)=0;
12            A(i,k+1:n)=A(i,k+1:n)-m*A(k,k+1:n); %for j=k+1:n
13            b(i)=b(i)-m*b(k);
14        end
15    end
16    deter=deter*A(n,n);
17    if A(n,n)==0
18        disp('Attenzione det(A)=0')
19    end
20 end

```

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Value

[2, 3, 4, 1, 6, 3, 3, 4, 8]

[3, 4, 32]

[3, 2.5000; -27.3333]

205

0

[2, -3, 4, 0, 7.5000, -5, 0, 0, 13, ...]

[4, -1, -2]

[4, -1, -2]

xvera =

4

-1

-2

errR =

0

```

1 A=[2 -3 4; 1 6 -3; -3 4 8]; b=[3; 4; -32];
2 [U,c,deter] = gaussNaiveOld(A,b);
3 x=backs(U,c)
4 xvera=[4;-1;-2]
5 errR=norm(x-xvera)/norm(xvera)

```